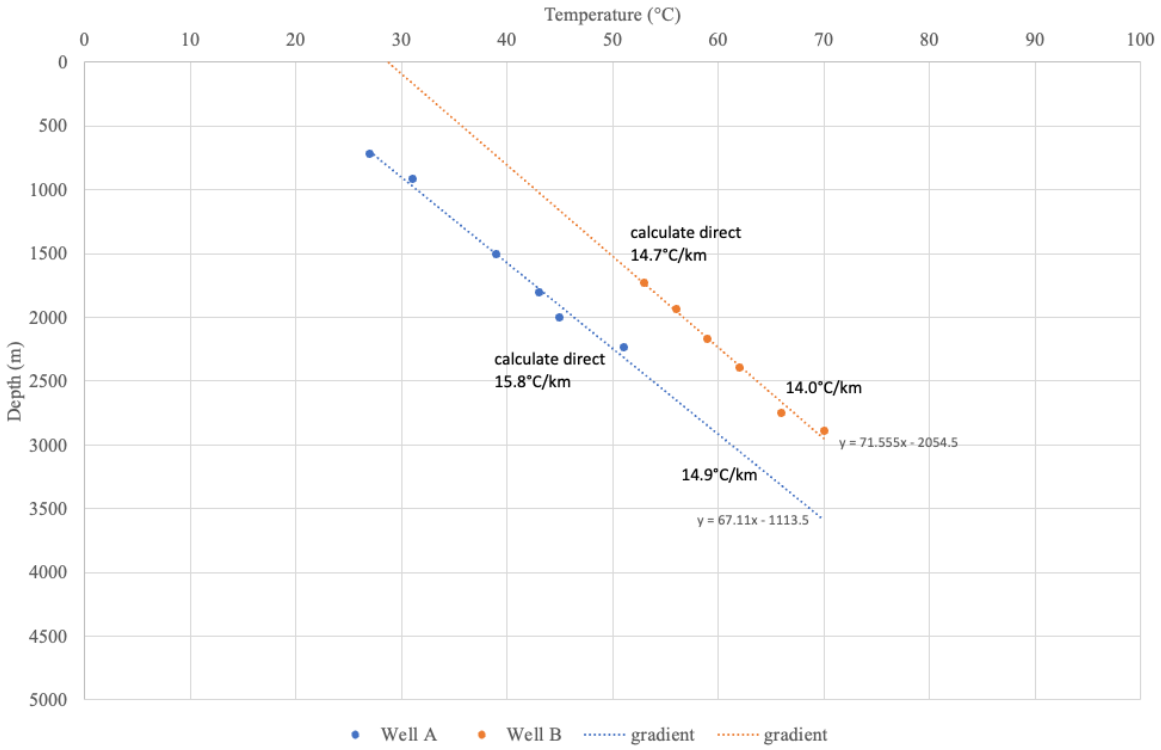
Practical 2 – temperature and pressure gradients

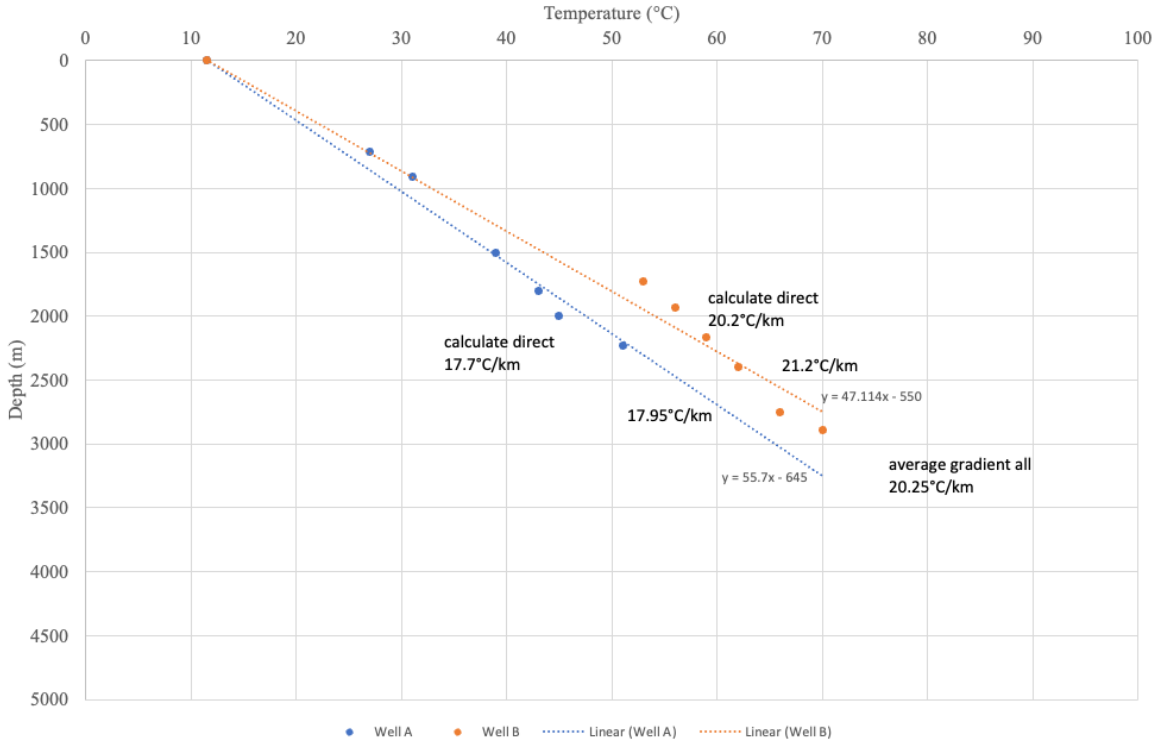
2.1 Calculation of temperature with depth (Data modified after Hirst 2015)
Temperature data have been acquired in 2 wells of the Cheshire Basin. Well locations are in figure 1, and data are in table 1.

1) Calculate a regional geothermal gradient for the Cheshire Basin based on wells A and B, and assuming an average surface temperature of 11.5°C. Exclude data for B1 for now.
Well A:
Calculate: 51-27 = 24°C / 1.515 km = 15.8°C/km
Graphically/Excel: 14.9°C/km
Well B:
Calculate: 70-53 = 17°C / 1.158 km = 14.7°C/km
Graphically/Excel: 14.0°C/km

Forcing through surface temperature of 11.5°C (based on literature):
Well A:
Calculate: 51-11.5 = 39.5°C / 2.23 km = 17.7°C/km
Graphically/Excel: 17.95°C/km
Well B:
Calculate: 70-11.5 = 58.5°C / 2.289 km = 20.2°C/km
Graphically/Excel: 21.2°C/km

Regional gradient: ~20.2°C

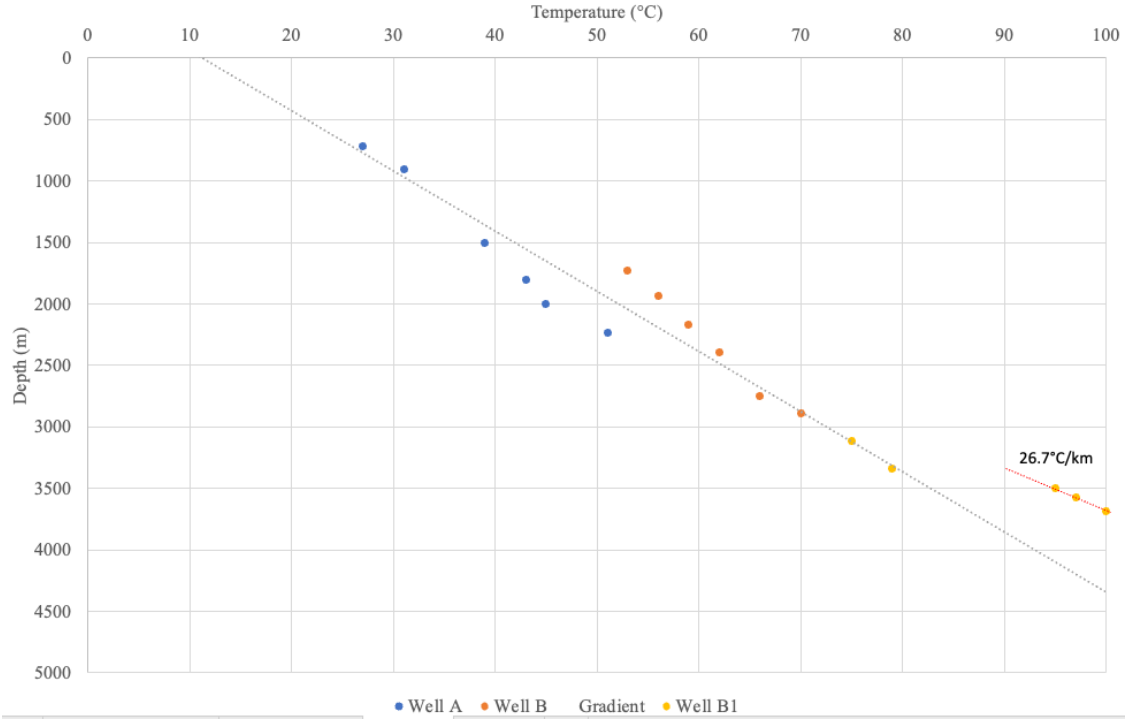




2) Later, an extension of well B was carried out by drilling deeper. Temperature data for this well B1 are equally in table 1. What are your observations on the new temperature data, and update your temperature gradient calculations where necessary.

Down to 3500m the data follow the same trend

Below 3500m: 100-95 = 5°C / 0.187 km = 26.7°C/km

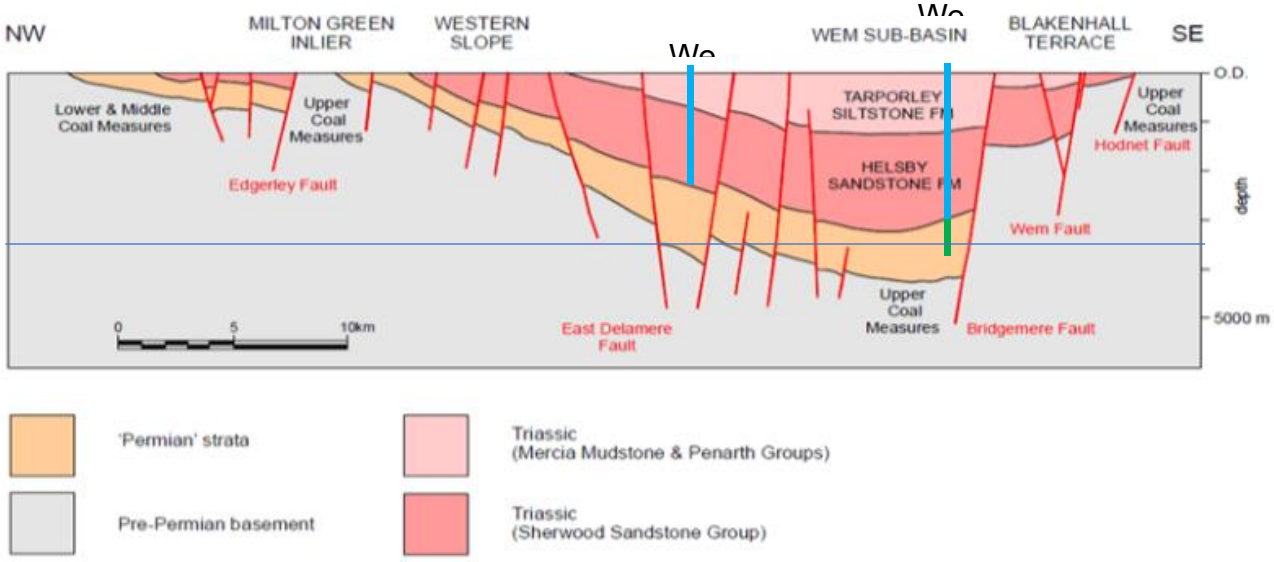


3) How has the geology changed in this well extension?

Increased geothermal gradient

Passed fault, could be basin bounding fault that changed dip

Increased heat flow along fault -> geothermal potential?



2.2 Calculation of pressure with depth

In the Cheshire Basin a gas well was drilled. The following formation pressures were measured downhole in the well. All depths are in m below ground level.

500m	58 kg/cm^2
1000m	120 kg/cm^2
1500m	175 kg/cm^2
2000m	300 kg/cm^2
2500m	400 kg/cm^2
3000m	450 kg/cm^2
3250m	500 kg/cm^2

Note that values are given in kg/cm^2, which is an old pressure unit, now largely superseded by the SI unit Pa(scal). It is still widely used in drilling though. This unit relates pressure with weight, which is useful in applications where a pressure is applied to generate a load, as in this case.

1 kg/cm^2 = 98,066.5 Pa.

1) Plot formation pressure data against depth.

2) Assuming an average rock density of 2.3g/cm^3, plot the lithostatic (overburden) pressure on the same graph. This is the pressure exerted by the weight of the overlying rock, so is the product of depth and rock density. The value of 2.3g/cm^3 is assumed to be the maximum overpressure that can be encountered before rupture.

Note 1: Mind the conversion!

Lithostatic pressure = Depth (m) x average lithostatic density (2.3g/cc))/10

The factor 10 is a conversion factor derived from units:

$$Density \times Depth\ in\ units: \frac{g}{cm^3} \times m = \frac{10^{-3}\ kg \times 100\ cm}{cm^3} = \frac{10^{-1}\ kg}{cm^2}$$

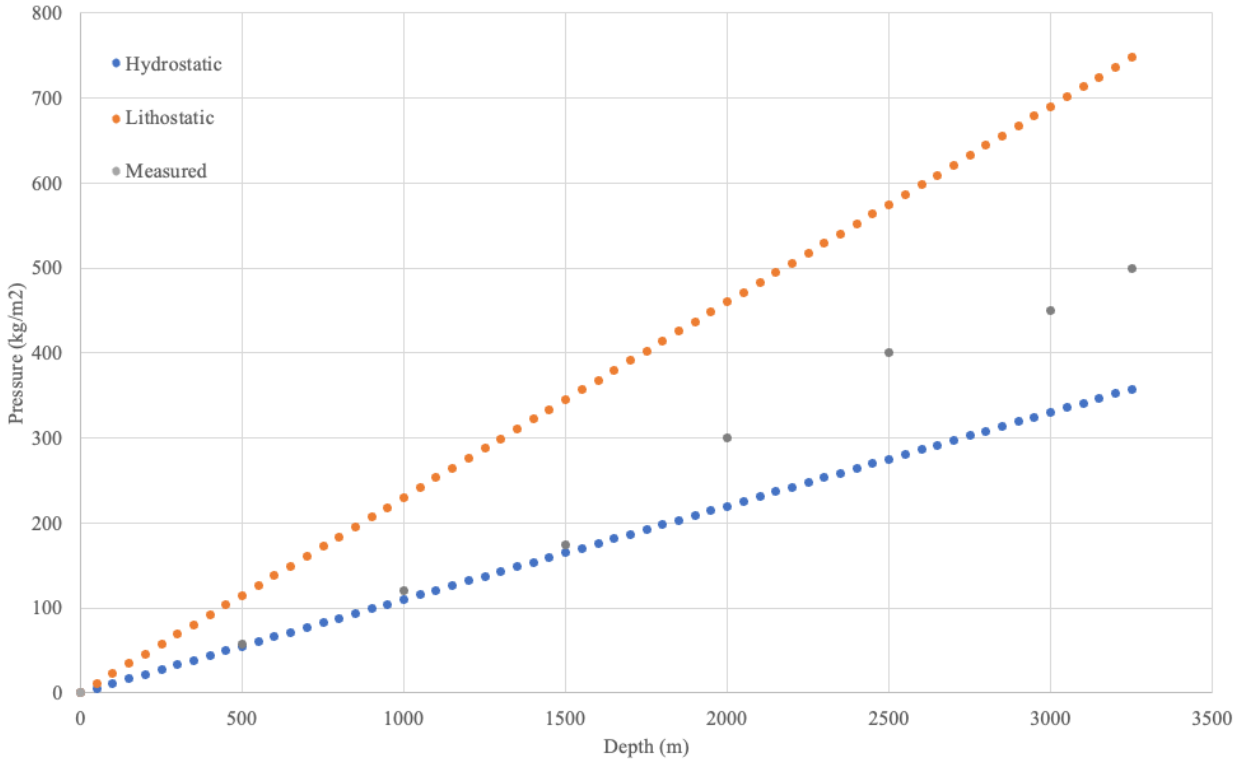
Calculate pressure values for range of depth in well

3) Calculate the hydrostatic pressure and plot it on the same graph.

Hydrostatic pressure = height of water column (m) x density of fluid (g/cm^{-3})/10

The factor 10 is a conversion factor

250m	(250*1.1)/10	27.5kg cm-2
300m	(300*1.1)/10	33kg cm-2
500m	(500*1.1)/10	55kg cm-2
1000m	(1000*1.1)/10	110kg cm-2
1500m	(1500*1.1)/10	165kg cm-2
2500m	(2500*1.1)/10	275kg cm-2
3250m	(3250*1.1)/10	357.5kg cm-2



4) What is the overpressure in the well, i.e. the difference between measured formation pressures and hydrostatic pressures? What does this tell us about conditions in the subsurface?

Overpressure = measured pressure – hydrostatic pressure

Note at 500m and 1000m KB, the measured pressure is only slightly above hydrostatic.

However, there is an in increase in pressure between 1500m and 2000m, perhaps due to:

a) impermeable layer that has prevented fluid expulsion (ie undercompaction)

b) charging by hydrocarbon, which will increase pore pressure